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1  package cdc;
2
3  import java.lang.String;
4  import java.util.*;
5  import javax.swing.*;
6  import java.awt.*;
7  import java.awt.event.*;
8
9  public class CDC
10 {
11     public String s, s1;
12     public String s2[][] = new String[20][20];
13     public VirtualCharacter Characters[] = new VirtualCharacter[4];
14     public int inputx[] = new int[2];
15     public int inputy[] = new int[2];
16     public int inputgold[] = new int[3];
17     public int inputsiliver[] = new int[3];
18     public int inputmoney[] = new int[5];
19     public int inputpick[] = new int[3];
20     public int inputmode[] = new int[1];
21     int players;
22     public boolean check[][] = new boolean[20][20];
23     public int map[][] = new int[20][20];
24     public int tempx, tempy, tempgold, tempsiliver, tempmoney, temppick;
25     public int i1, i2, i3, i4, i5, i6, i7, i8;
26     public int judge = 0;
27     public String temp;
28     public String item = "x0x0";
29
30     Vector v = new Vector();
31
32     public CDC()
33     {
34         s = "8"; //判斷字元
35         i1 = i2 = i3 = i4 = i5 = i6 = i7 = i8 = 0;
36         players = 0; //玩家人數
37         for (int j = 0; j < 20; j++) //initialize 所有物品
38         {
39             for (int k = 0; k < 20; k++)
40             {
41                 check[j][k] = true; //預設true可挖false已挖
42                 int rand;
43                 rand = (int) (Math.random() * 8) + 1; //rand = 1~8共八項物品
44                 if ( rand == 1 && i1 > 171 ) //限制金礦個數 = 171
45                 {
46                     k--;
47                     continue;
48                 }
49                 else if ( rand == 1 && i1 <= 171 )
50                     i1++;
51                 if ( rand == 2 && i2 > 200 ) //限制銀礦個數 = 200
52                 {
53                     k--;
54                     continue;
55                 }
56                 else if ( rand == 2 && i2 <= 200 )
57                     i2++;
58                 if ( rand == 3 && i3 > 5 ) //限制+2金個數 = 5
59                 {
60                     k--;
61                     continue;
62                 }
63                 else if ( rand == 3 && i3 <= 5 )
64                     i3++;
65                 if ( rand == 4 && i4 > 4 ) //限制瞬間移動個數 = 4
66                 {
67                     k--;
68                     continue;
69                 }
70                 else if ( rand == 4 && i4 <= 4 )
71                     i4++;
72                 if ( rand == 5 && i5 > 5 ) //限制-2金個數 = 4
73                 {

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74         k--;
75         continue;
76     }
77     else if ( rand == 5 && i5 <= 5 )
78         i5++;
79     if ( rand == 6 && i6 > 8 )//限制捶子個數 = 8
80     {
81         k--;
82         continue;
83     }
84     else if ( rand == 6 && i6 <= 8 )
85         i6++;
86     if ( rand == 7 && i7 > 8 )//限制金礦個數 = 171
87     {
88         k--;
89         continue;
90     }
91     else if ( rand == 7 && i7 <= 8 )
92         i7++;
93     if ( rand == 8 && i3 > 5 )//限制均富個數 = 8
94     {
95         k--;
96         continue;
97     }
98     else if ( rand == 8 && i8 <= 5 )
99         i8++;
100     map[j][k] = rand;
101 }
102 }
103 check[0][0] = false;
104 check[0][1] = false;
105 check[1][0] = false;
106 check[19][0] = false;
107 check[19][1] = false;
108 check[18][0] = false;
109 check[0][19] = false;
110 check[1][19] = false;
111 check[0][18] = false;
112 check[19][19] = false;
113 check[18][19] = false;
114 check[19][18] = false;
115 check[9][9] = false;
116 check[9][10] = false;
117 check[10][9] = false;
118 check[10][10] = false;
119 map[0][0] = 0;
120 map[0][1] = 0;
121 map[1][0] = 0;
122 map[19][0] = 0;
123 map[19][1] = 0;
124 map[18][0] = 0;
125 map[0][19] = 0;
126 map[1][19] = 0;
127 map[0][18] = 0;
128 map[19][19] = 0;
129 map[18][19] = 0;
130 map[19][18] = 0;
131 map[9][9] = 0;
132 map[9][10] = 0;
133 map[10][9] = 0;
134 map[10][10] = 0;
135 }
136
137 public Vector update()
138 {
139     s1 = "R";
140     temp = "";
141     if (judge == 0)
142     {
143         v.clear();
144         v.addElement(s);
145     }
146     else if (judge == 1)

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147 {
148     for (int i = 0; i < 4; i++)
149     {
150         tempx = Characters[i].x;
151         tempy = Characters[i].y;
152         tempgold = Characters[i].gold;
153         tempsilver = Characters[i].silver;
154         temppick = Characters[i].pick;
155         tempmoney = Characters[i].money;
156         if (Characters[i] != null)
157         {
158             for (int p = 1; p >= 0; p--)
159             {
160                 inputx[p] = Characters[i].x % 10;
161                 inputy[p] = Characters[i].y % 10;
162                 Characters[i].x /= 10;
163                 Characters[i].y /= 10;
164             }
165             for (int q = 2; q >= 0; q--)
166             {
167                 inputgold[q] = Characters[i].gold % 10;
168                 inputgold[q] = Characters[i].silver % 10;
169                 inputpick[q] = Characters[i].pick % 10;
170                 Characters[i].gold /= 10;
171                 Characters[i].silver /= 10;
172                 Characters[i].pick /= 10;
173             }
174             for (int t = 4; t >= 0; t--)
175             {
176                 inputmoney[t] = Characters[i].money % 10;
177                 Characters[i].money /= 10;
178             }
179             inputmode[0] = Characters[i].mode;
180             s1=s1.valueOf(s1+(i+1)+Characters[i].pic+inputx[0]+inputx[1]+
181                 inputy[0]+inputy[1]+Characters[i].dir+inputgold[0]+
182                 inputgold[1]+inputgold[2]+inputsilver[0]+
183                 inputsilver[1]+inputsilver[2]+inputmoney[0]+
184                 inputmoney[1]+inputmoney[2]+inputmoney[3]+inputmoney[4]+
185                 inputpick[0]+inputpick[1]+inputpick[2]+inputmode[0]);
186         }
187         Characters[i].x = tempx;
188         Characters[i].y = tempy;
189         Characters[i].gold = tempgold;
190         Characters[i].silver = tempsilver;
191         Characters[i].money = tempmoney;
192         Characters[i].pick = temppick;
193     }
194     s1 = s1.valueOf(s1 + "M" + item);
195     v.clear();
196     v.addElement(s1);
197 }
198 if (judge == 0 && players >= 4 )
199 {
200     judge = 1;
201 }
202 return v;
203 }
204 public synchronized void addVirtualCharacter(int clientno, int pic)
205 {
206     players++;
207     Characters[clientno - 1] = new VirtualCharacter();
208     Characters[clientno - 1].clientno = clientno;
209     Characters[clientno - 1].pic = pic;
210     Characters[clientno - 1].money = 0;
211     Characters[clientno - 1].gold = 0;
212     Characters[clientno - 1].silver = 0;
213     Characters[clientno - 1].pick = 1;
214     Characters[clientno - 1].mode = 0;
215     switch (clientno)
216     {
217     case 1:
218         Characters[clientno - 1].dir = 0;
219         Characters[clientno - 1].x = 0;

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220     Characters[clientno - 1].y = 0;
221     s = s.valueOf("C1" + Characters[clientno - 1].pic);
222     break;
223 case 2:
224     Characters[clientno - 1].dir = 0;
225     Characters[clientno - 1].x = 19;
226     Characters[clientno - 1].y = 0;
227     s = s.valueOf("C2" + Characters[clientno - 1].pic);
228     break;
229 case 3:
230     Characters[clientno - 1].dir = 0;
231     Characters[clientno - 1].x = 0;
232     Characters[clientno - 1].y = 19;
233     s = s.valueOf("C3" + Characters[clientno - 1].pic);
234     break;
235 case 4:
236     Characters[clientno - 1].dir = 0;
237     Characters[clientno - 1].x = 19;
238     Characters[clientno - 1].y = 19;
239     s = s.valueOf("C4" + Characters[clientno - 1].pic);
240     break;
241 }
242 }
243 public synchronized void updateDirection(int clientno, int MoveCode)
244 {
245     if (MoveCode == 1 && Characters[clientno - 1].y - 1 >= 0)
246     { //up and walk
247         Characters[clientno - 1].y = Characters[clientno - 1].y - 1;
248         System.out.println("第"+clientno+"位玩家方向向上走");
249         System.out.println("第"+clientno+"位玩家的x座標為"+Characters[clientno - 1].x);
250         System.out.println("第"+clientno+"位玩家的y座標為"+Characters[clientno - 1].y);
251         Characters[clientno - 1].dir = 1;
252     }
253     else if (MoveCode == 0 && Characters[clientno - 1].y + 1 <= 19)
254     { //down and walk
255         Characters[clientno - 1].y = Characters[clientno - 1].y + 1;
256         System.out.println("第"+clientno+"位玩家方向向下走");
257         System.out.println("第"+clientno+"位玩家的x座標為"+Characters[clientno - 1].x);
258         System.out.println("第"+clientno+"位玩家的y座標為"+Characters[clientno - 1].y);
259         Characters[clientno - 1].dir = 0;
260     }
261     else if (MoveCode == 2 && Characters[clientno - 1].x - 1 >= 0)
262     { //left and walk
263         Characters[clientno - 1].x = Characters[clientno - 1].x - 1;
264         System.out.println("第"+clientno+"位玩家方向向左走");
265         System.out.println("第"+clientno+"位玩家的x座標為"+Characters[clientno - 1].x);
266         System.out.println("第"+clientno+"位玩家的y座標為"+Characters[clientno - 1].y);
267         Characters[clientno - 1].dir = 2;
268     }
269     else if (MoveCode == 3 && Characters[clientno - 1].x + 1 <=19)
270     { //right and walk
271         Characters[clientno - 1].x = Characters[clientno - 1].x + 1;
272         System.out.println("第"+clientno+"位玩家方向向右走");
273         System.out.println("第"+clientno+"位玩家的x座標為"+Characters[clientno - 1].x);
274         System.out.println("第"+clientno+"位玩家的y座標為"+Characters[clientno - 1].y);
275         Characters[clientno - 1].dir = 3;
276     }
277     else if (MoveCode == 6 )//up and turn
278     {
279         if ( Characters[clientno - 1].y != 0 )
280
281             System.out.println("第"+clientno+"位玩家的上方物品屬性"+check[Characters[clientno - 1].x][Characters[clientno - 1].y -1]);
282             Characters[clientno - 1].dir = 1;
283             System.out.println("第"+clientno+"位玩家方向轉向上");
284             System.out.println("第"+clientno+"位玩家的x座標為"+Characters[clientno - 1].x);
285             System.out.println("第"+clientno+"位玩家的y座標為"+Characters[clientno - 1].y);
286         }
287     else if (MoveCode == 7 )//down and turn
288     {
289         if ( Characters[clientno - 1].y != 19 )
290
291             System.out.println("第"+clientno+"位玩家的下方物品屬性"+check[Characters[clientno - 1].x][Characters[clientno - 1].y +1]);
292             Characters[clientno - 1].dir = 0;
293             System.out.println("第"+clientno+"位玩家方向轉向下");
294             System.out.println("第"+clientno+"位玩家的x座標為"+Characters[clientno - 1].x);
295             System.out.println("第"+clientno+"位玩家的y座標為"+Characters[clientno - 1].y);
296         }
297     }
298 }

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        tno - 1].x][Characters[clientno - 1].y + 1]);
290 Characters[clientno - 1].dir = 0;
291 System.out.println("第"+clientno+"位玩家方向轉向下");
292 System.out.println("第"+clientno+"位玩家的x座標為"+Characters[clientno - 1].x);
293 System.out.println("第"+clientno+"位玩家的y座標為"+Characters[clientno - 1].y);
294 }
295 else if (MoveCode == 8 )//left
296 {
297     if (Characters[clientno - 1].x != 0 )
298
299         System.out.println("第"+clientno+"位玩家的左方物品屬性"+check[Characters[clientno - 1].x-1][Characters[clientno - 1].y ]);
300 Characters[clientno - 1].dir = 2;
301 System.out.println("第"+clientno+"位玩家方向轉向左");
302 System.out.println("第"+clientno+"位玩家的x座標為"+Characters[clientno - 1].x);
303 System.out.println("第"+clientno+"位玩家的y座標為"+Characters[clientno - 1].y);
304 }
305 else if (MoveCode == 9 )//right
306 {
307     if (Characters[clientno - 1].x != 19)
308
309         System.out.println("第"+clientno+"位玩家的右方物品屬性"+check[Characters[clientno - 1].x+1][Characters[clientno - 1].y ]);
310 Characters[clientno - 1].dir = 3;
311 System.out.println("第"+clientno+"位玩家方向轉向右");
312 System.out.println("第"+clientno+"位玩家的x座標為"+Characters[clientno - 1].x);
313 System.out.println("第"+clientno+"位玩家的y座標為"+Characters[clientno - 1].y);
314 }
315 }
316 public synchronized int getItem(int clientno)
317 {
318     int tmpx,tmpy;
319     int index = 0;
320     int ix = 0;
321     int iy = 0;
322     System.out.println("第"+clientno+"位玩家在挖石頭");
323     System.out.println("第"+clientno+"位玩家的方向為"+Characters[clientno-1].dir);
324     System.out.println("第"+clientno+"位玩家的x座標為"+Characters[clientno - 1].x);
325     System.out.println("第"+clientno+"位玩家的y座標為"+Characters[clientno - 1].y);
326     if (Characters[clientno - 1].dir == 2 && Characters[clientno - 1].x - 1 >= 0)
327         //left
328     {
329         ix = Characters[clientno - 1].x - 1;
330         iy = Characters[clientno - 1].y;
331     }
332     else if (Characters[clientno - 1].dir == 0 && Characters[clientno - 1].y + 1 <= 19) //down
333     {
334         ix = Characters[clientno - 1].x;
335         iy = Characters[clientno - 1].y + 1;
336     }
337     else if (Characters[clientno - 1].dir == 3 && Characters[clientno - 1].x + 1 <= 19) //right
338     {
339         ix = Characters[clientno - 1].x + 1;
340         iy = Characters[clientno - 1].y;
341     }
342     else if ( Characters[clientno - 1].dir == 1 && Characters[clientno - 1].y - 1 >= 0)//up
343     {
344         ix = Characters[clientno - 1].x;
345         iy = Characters[clientno - 1].y - 1;
346     }
347     System.out.println("第"+clientno+"位玩家挖的x座標為"+ix);
348     System.out.println("第"+clientno+"位玩家挖的y座標為"+iy);
349     System.out.println("第"+clientno+"位玩家挖的地點為"+check[ix][iy]);
350     System.out.println("第"+clientno+"位玩家挖的東西為"+map[ix][iy]);
351     check[ix][iy] = false;
352     item = item.valueOf( item + "x"+ix + "x"+iy );
353     if (map[ix][iy] == 1)
354     {
355         //1 金礦
356         index = map[ix][iy];
357         System.out.println("第"+clientno+"位玩家挖到金礦");
358     }
359 }

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354     Characters[clientno - 1].money += 50 + Characters[clientno - 1].pick*(int)
      (Math.random() * 50);
355     Characters[clientno - 1].gold += Characters[clientno - 1].gold;
356     map[ix][iy] = 0;
357 }
358 else if (map[ix][iy] == 2)
359 {
360     //2 銀礦
361     index = map[ix][iy];
362     System.out.println("第"+clientno+"位玩家挖到銀礦");
363     Characters[clientno - 1].money += Characters[clientno - 1].pick*(int)
      (Math.random() * 50);
364     Characters[clientno - 1].siliver += Characters[clientno - 1].siliver;
365     map[ix][iy] = 0;
366 }
367 else if (map[ix][iy] == 3)
368 {
369     //3 金錢兩倍
370     index = map[ix][iy];
371     System.out.println("第"+clientno+"位玩家挖到+2金");
372     Characters[clientno - 1].money *= 2;
373     map[ix][iy] = 0;
374 }
375 else if (map[ix][iy] == 4) //4 隨機傳送至一空格處
376 {
377     index = map[ix][iy];
378     System.out.println("第"+clientno+"位玩家挖到move");
379     map[ix][iy] = 0;
380     do
381     {
382         tmpx = (int) (Math.random()*16)+2;
383         tmpy = (int) (Math.random()*16)+2;
384     }
385     while(check[tmpx][tmpy]);
386     Characters[clientno -1].x = tmpx;
387     Characters[clientno -1].y = tmpy;
388     System.out.println("第"+clientno+"位玩家傳到"+tmpx+" "+tmpy);
389 }
390 else if (map[ix][iy] == 5) //5 金錢減半
391 {
392     index = map[ix][iy];
393     Characters[clientno - 1].money /= 2;
394     System.out.println("第"+clientno+"位玩家挖到-2金");
395     map[ix][iy] = 0;
396 }
397 else if (map[ix][iy] == 6) //6 鎚子升級
398 {
399     index = map[ix][iy];
400     System.out.println("第"+clientno+"位玩家挖到pick");
401     Characters[clientno -1].pick++;
402     map[ix][iy] = 0;
403 }
404 else if (map[ix][iy] == 7)
405 {
406     //7 控制顛倒
407     index = map[ix][iy];
408     System.out.println("第"+clientno+"位玩家挖到島");
409     if(Characters[clientno -1].mode == 1)
410     {
411         Characters[clientno - 1].mode--;
412         map[ix][iy] = 0;
413     }
414     else
415     {
416         Characters[clientno - 1].mode++;
417         map[ix][iy] = 0;
418     }
419 }
420 else if (map[ix][iy] == 8) //8 均富
421 {
422     index = map[ix][iy];
423     System.out.println("第"+clientno+"位玩家挖到=");
424     map[ix][iy] = 0;
425     int tmp = 0;
426     for(int j=0;j<4;j++)
427     {

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425         tmp += Characters[j].money;
426     }
427     tmp /= 4;
428     for(int j=0;j<4;j++)
429     {
430         Characters[j].money = tmp;
431     }
432 }
433 System.out.println("第1位玩家錢數"+Characters[0].money);
434 System.out.println("第2位玩家錢數"+Characters[1].money);
435 System.out.println("第3位玩家錢數"+Characters[2].money);
436 System.out.println("第4位玩家錢數"+Characters[3].money);
437
438 return index;
439 }
440 }
```